

Pretreatment transcriptional state carries condition-specific information about future clonal detection in a lineage-traced melanoma line

CellFate-Rx, Generation 1.

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evidence lock	60602531449079b8f86debea9ffd69753f8bfad033be4c476751d39da08c78aa
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Both digests are verifiable from the repository. See **Availability of data and materials**.

Abstract

Background. Whether a cell's molecular state before a perturbation predicts what happens to it afterwards is usually asked retrospectively, after outcome and state have been measured in the same cells. We ask it prospectively at clone level, in one BRAF-V600E melanoma cell line (WM989, GSE279162), in which a barcoded population was split across six observed experimental conditions: Acid, Cisplatin, CoCl₂, Dabrafenib, Doxorubicin and Trametinib. Of the clones that experiment recovered, 1,401 carry a pretreatment profile and are therefore analysable prospectively; those are the clones used here. The ranking test uses the 892 of them detected under at least one condition and undetected under at least one other.

Results. Within this system, pretreatment gene expression contains condition-specific information about future clonal detection beyond condition identity and captured pretreatment clone abundance, under clone-held-out evaluation whose folds, features and exclusions were fixed before any model was fitted. Under a test preregistered in full — the metric, population, weighting, comparator, null and verdict rule all fixed by digest before any ranking statistic was computed, though after earlier predictive analyses of the same data — a frozen state-by-condition interaction model improves clone-specific ordering of the six conditions over a non-interactive additive model: +0.051605 in equal-clone-weighted within-clone AUROC, 95% CI [+0.037197, +0.065571], with 0 of 1000 full-refit permutation draws reaching the observed value ($p < 0.001$).

Conclusions. A state contribution shared additively across all conditions cannot change their ordering within a clone. Allowing state effects to vary by condition improved that ordering in WM989: part of what a clone's pretreatment state carries bears on which of the six conditions it is still detected after, not only on how detectable it is overall. The outcome is a detection proxy and is not death, sensitivity, resistance or clinical response. The design and the artifacts behind it are released, so that the result can be reproduced from the public data and tested in other lineage-traced systems, where it has not yet been tested.

Keywords

clonal barcoding; lineage tracing; drug resistance; preregistration; reproducibility; within-clone ranking; permutation test; melanoma; reanalysis

Background

The question is easy to state and hard to evaluate honestly: before a perturbation is applied, does a cell's transcriptional state say anything about which perturbation that cell is still detected after?

Barcoded lineage tracing solves the hardest part of asking it. A clone is split, part of it is profiled before anything is done to it, and the remaining parts are exposed to different conditions, so the pretreatment profile is genuinely prior to the outcome rather than a consequence of it.

That design, and the data used here, are not ours. Schaff et al. built exactly such a system across six conditions in parallel and showed that pre-existing state predicts which clones go on to resist, identifying high CD44 expression in treatment-naïve cells as a marker of resistance across multiple conditions [1]. This work is a reanalysis of their data and takes no credit for the experiment.

What that establishes is a **general propensity**: some clones resist more of the conditions than others do. It leaves a different question open — whether pretreatment state says *which* condition a given clone is still detected after, rather than how resistant that clone is overall. The two are separable, and separating them is what this work does.

Two things make the ordering question hard to evaluate rather than merely to observe. The first is that clone abundance dominates: a clone that was large before treatment is more likely to be detected after it for reasons that have nothing to do with state, so any comparison that does not hold abundance fixed will find a signal that is really a headcount. The second is that clone-specific ordering is not something a model can produce by being generally right about a clone — it requires

an explicit interaction between state and condition, because a state contribution shared additively across the six conditions shifts all six of a clone's scores together and leaves their order untouched.

This work evaluates exactly that, once, under a protocol frozen in advance.

Relation to prior work

The finding that pre-existing, non-genetic single-cell state predicts which cells resist therapy is established in this system and is not claimed here. Shaffer et al. showed that rare transcriptional states in WM989 predict which cells resist vemurafenib and are stabilised by drug exposure [3]; Emert et al. resolved substructure within those rare states and linked it to distinct resistant outcomes [4]; Goyal et al. showed that clonal fates after drug are largely predetermined by pre-treatment molecular differences and are diverse rather than binary [5]; and Schaff et al. extended clonal tracing to six conditions in parallel, reporting cross-condition resistance correlation and CD44 as a marker of resistance across several of them [1].

We do not claim to be the first to look at condition-specific expression in this system. Schaff et al.'s own deposited analysis reports condition-associated markers and signatures alongside the cross-condition results, and earlier work resolves substructure that distinguishes resistant outcomes [4,5]. What has not been done is the specific thing tested here: a **frozen, clone-held-out, preregistered test of whether an explicit state-by-condition interaction improves clone-specific ORDERING** over a non-interactive additive model, with captured pretreatment abundance held fixed in every model including the null.

The distinction the design turns on is not rhetorical. A state contribution shared additively across the six conditions — including whatever part of a general resistance propensity, or of a per-clone marker level, is the same for each of them — shifts all six of that clone's predicted scores together and therefore contributes **exactly zero** to a within-clone ordering metric. The comparator was chosen to enforce that separation before any ranking statistic was computed, and the outcome is that the additive state term contributes nothing to ordering while the interaction contributes all of the gain. The contribution is the evaluation and its preregistration, not the observation that state carries condition-relevant information.

The methodological posture is borrowed rather than invented. Kapoor and Narayanan catalogue eight kinds of leakage across 294 papers in seventeen fields, and observe that complex models frequently fail to beat logistic regression once the leakage is corrected [6]. That is the failure mode this design is built against: the comparator is a simpler model of the same family, every preprocessing step is refitted inside the training fold, the permutation null refits the whole pipeline rather than shuffling labels, and the metric, population, comparator and verdict rule were fixed in a digest-frozen protocol before any ranking statistic was computed. That

protocol was written after, and in the light of, earlier predictive analyses of the same data, which the stage records keep in full.

Methods

Data

ROLE B, primary	GSE279162 (WM989)	benchmark, model, tool and ranking analysis
ROLE A, supporting	GSE227151 (Rewind)	historical supporting evidence only

Role B is one BRAF-V600E melanoma cell line. The source experiment recovered many thousands of barcoded clones; **1,401 of them carry a pretreatment observation**, which is what a prospective clone-level question requires, and those 1,401 are the analysis population here. It carries the whole primary claim.

Role B is a reanalysis. We generated no new data. The experiment was performed by Schaff et al. [1], who introduced a barcode library into WM989 A6-G3, isolated 350,000 uniquely barcoded cells, expanded them for approximately six doublings, and divided the population across twelve treatment arms — two replicates of each of the six conditions reanalysed here. Dabrafenib, trametinib, CoCl₂ and acidic media were applied continuously; cisplatin and doxorubicin were applied for a treatment period followed by a recovery period, each arm spanning four weeks in total. Exact concentrations and schedules are in [1].

Role A is a separately reconstructed reprogramming system [2]. It contributes one supporting sentence and nothing else; its own confirmation gate failed. It is not a replication of Role B, and it does not provide the same multi-condition task or the same outcome.

No additional dataset was searched, downloaded, qualified or used. Raw sequencing data is not vendored; accessions are given above. **Figure 1** summarises the design and the evaluable population.

Benchmark construction

Clone assignments, outcome construction, the five outer folds, feature rules, condition aliases and exclusions were all fixed before any model was fitted, and were not revised afterwards.

Two endpoint families were built. **C1** is post-treatment clone detection: an observed zero means no assigned post-treatment cell was seen for that clone-condition row. **C2** is a clone-balanced abundance endpoint. Everything reported below is C1.

Expression is clone-level pseudobulk: raw pretreatment counts are summed over a clone's cells, then CP10K-normalised and log_{1p}-transformed exactly once. Applying

the transform twice, or summing already-normalised cells, produces a different feature space and a model the benchmark never evaluated.

Models

W1	$B + U$	nuisance and condition identity only
W4	$X + B + U$	plus an additive expression term
W5	$X + B + U + X*U$	plus an explicit state-by-condition interaction

x is the clone expression profile reduced to 50 principal components on a train-only basis. B is the captured-abundance nuisance block: $\log_{10}$ cell counts, total and per pretreatment library. U is condition identity, encoded as five non-reference indicators with Acid as reference. The design has 309 columns: 50 components, 4 nuisance terms, 5 indicators and 250 interaction terms.

B is not optional. Abundance is the confounder that makes a naive version of this question trivial, so it is in every model including the null, and the tool refuses to score without it.

Evaluation

Five outer folds, held out **by clone**, so no model ever scores a clone it trained on. Gene filtering, the PCA basis and every scaler are refitted inside each training fold. Hyperparameters are selected by an inner grouped split within the training folds only.

The preregistered ranking test

The primary question is whether W5 orders the six conditions *within a clone* better than W4. The metric is equal-clone-weighted within-clone AUROC: for each clone, the mean over positive/zero condition pairs with ties scoring exactly 0.5; then a plain mean over clones, so a clone contributing many pairs cannot outweigh one contributing few.

The evaluable population is clones with at least one detected and one undetected condition; a clone with no contrast has an undefined within-clone AUROC. This yields 892 of 1,401 clones — 472 were never detected under any condition and 37 were always detected.

The comparator is W4, not W1. An additive expression term cannot by construction create clone-specific ordering, so W4 isolates the interaction as the only thing that could.

The null is a **full refit**: profiles are permuted within stratum on each side of the outer-fold boundary and never across it, and the entire pipeline — filtering, PCA, scaler fitting and hyperparameter selection — is re-run inside every draw. Observed-data hyperparameters are never reused. 1000 draws, no early stopping, with a completeness assertion that refuses an incomplete null rather than silently reporting a smaller one.

Uncertainty on the observed statistic is a 2,000-replicate clone bootstrap, conditional on the fitted models.

AI use disclosure

Generative AI assistants were used throughout this work. Claude Opus 5 and Claude Opus 4.8 (Anthropic) wrote most of the code and documentation. Gemini 3.1 Pro (Google), Claude Haiku 4.5 (Anthropic), GPT 5.6 sol and GPT 6 astra (OpenAI) were used to check that work. The study’s logic and direction came mainly from the author. Each protocol was fixed by cryptographic digest before the statistics it governs were computed, and every number reported here is produced by the frozen code and traced mechanically to a locked artifact. The author reviewed the work and takes full responsibility for its content.

Results

The interaction improves clone-specific ordering

R(W1)	0.692654	nuisance + condition
R(W4)	0.692176	+ additive X
R(W5)	0.743781	+ explicit X x U

delta_RANK	+0.051605	CI95 [+0.037197, +0.065571]
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Figure 2 shows the three models and the observed statistic against its null.

R(W4) sits *below* R(W1) by 0.0005. The additive expression term contributes nothing to ordering, which is precisely why W4 was preregistered as the comparator. The entire ordering gain is the interaction.

This also says what the gain is not. A state contribution shared additively across all six conditions cannot change their ordering within a clone: it moves all six of that clone’s scores together, and within-clone AUROC compares those six scores, so such a contribution counts **exactly zero** here by construction. W4 carries exactly that contribution, and it adds nothing to the ordering. Allowing state effects to vary by condition, as W5 does, is what improved it. That does not rule out one shared biological programme behind those effects: a multi-condition signal of the kind CD44 marks in this system [1] would be such a programme, and one whose influence differs between conditions would appear as an interaction too. What this measurement locates is condition-dependent state effects, not the number of programmes behind them.

The separation, not the p-value, is the result

null p95	0.008672
observed / null p95	6.0x
observed vs null mean	11.8 null standard deviations above it
largest of 1,000 draws	0.013722 -- the observed value exceeds EVERY null draw
draws reaching observed	0 of 1000 full-refit permutations

$p < 0.001$ is the floor of a 1,000-draw permutation test and is reported as such, never as a point estimate. The number that carries weight is the separation: nothing the null produced came close.

It holds in every stratum it was broken down by

BY OUTER FOLD		BY PRETREATMENT DEPTH	
fold 0	+0.0435	1 cell	+0.0535
fold 1	+0.0548	2 cells	+0.0528
fold 2	+0.0658	3-4	+0.0314
fold 3	+0.0506	5-9	+0.0462
fold 4	+0.0435	10+	+0.0779

Positive in all five folds and all five depth strata (**Figure 3A, 3B**). These were preregistered as descriptive and could not have rescued a failed primary gate; they were not asked to. The breakdown is by fold and by depth only; across the six conditions the interaction is not uniform (Limitation 3).

Choosing the lowest-scoring condition

delta_TOP1 +0.115471 CI95 [+0.082960, +0.145740]

Selecting each clone's lowest predicted detection score finds a genuine zero for 82.8% of evaluable clones under W5 against 71.3% under W4 (**Figure 3C**). This was preregistered as a directional-consistency check, not a significance test: it could withhold support, never grant it. It did not withhold.

The tool

A frozen predictor ships with this work. For one starting clone it returns a `future_detection_score` for each of the six observed conditions, reproducing the frozen out-of-fold predictions to within 5e-16.

What it refuses is as much of the specification as what it returns:

an unknown condition	-> UNSUPPORTED_TREATMENT, and no score
a missing nuisance block	-> MISSING_REQUIRED_NUISANCE, never imputed
a wrong feature schema	-> UNSUPPORTED_FEATURE_SCHEMA
a validated ordering	-> withheld unless the preregistered verdict file is supplied

The refusal was tested adversarially rather than asserted. 56 of 56 hostile condition strings were refused: case and whitespace variants, dose formats, unicode look-alikes, controls, and sixteen real oncology drugs including Vemurafenib — the drug for this exact mutation — and Carboplatin, one substitution from a condition that *is* supported.

That test found something worth stating. Acid is the reference level and is encoded as five zero indicators, so the indicator encoder alone maps *any* unrecognised string to the Acid row and would return the Acid score under another name. The vocabulary filter is the only thing preventing this, and it was verified to hold rather than assumed to.

Discussion

The result is narrow by design, and this section keeps apart three things that are easy to run together: what the result cannot support, what may not be said about it in any form, and what would actually test it. The first two are separate on purpose; neither should have to be inferred from the other.

Limitations

Carried verbatim from the preregistered verdict.

- 1 No independent biological replication of the Role-B finding. Clone-held-out folds and two endpoint families are not replication.
- 2 Captured pretreatment clone abundance remains $\sim 3.45\times$ the whole state contribution. The ordering is abundance first, then condition-specific state.
- 3 Four of six conditions carry meaningful interaction. Cisplatin is negligible on C1 and Doxorubicin is negative on both endpoint families.
- 4 C1 is an observed detection proxy -- not death, sensitivity or clinical response.
- 5 The bootstrap interval is conditional on the fitted models; only the null refits.
- 6 Role A remains positive-but-underpowered supporting evidence. Its confirmation gate 18.3 FAILED at 0.64 against a 0.80 threshold, and a later audit of the instrument put the true power at 0.45 -- lower still. Its effect size must not be quoted as an estimate, because an underpowered design that reaches significance inflates it.

Limitation 2 is the one most likely to be misread. This work does not show that state dominates outcome. It shows that state adds something specific, on top of an abundance term that is several times larger.

Limitation 3 has a pattern that this experiment cannot interpret. Schaff et al. chose the six conditions as three pairs [1]: two targeted inhibitors, dabrafenib against BRAF and trametinib against MEK; two non-clinical stresses, CoCl₂ mimicking hypoxia and acidic media mimicking acidosis; and two DNA-damaging agents, cisplatin cross-linking DNA and doxorubicin inhibiting topoisomerases. The four conditions that carry meaningful interaction are the first two pairs, and the two that do not are the DNA-damaging pair. The same four were also the ones applied continuously, and the same two were given for a treatment period followed by recovery (Methods), so class and treatment schedule divide the six conditions identically and cannot be told apart here. With two conditions in each class, and a grouping noticed only after the results, neither reading is a finding and neither is claimed. Separating them needs conditions in which class and schedule vary independently.

Limitation 6 is the honest position on Role A. Its own gate failed, we audited our own power calculation and found it had been too generous, and we report the worse number.

What this does not show

Separate from the limitations above, these are claims this work may not make, in any form.

- 1 NEVER unseen-condition generalization
- 2 NEVER cross-cell-line or cross-patient generalization
- 3 NEVER clinical treatment recommendation
- 4 NEVER causal treatment-effect estimation
- 5 NEVER a calibrated probability
- 6 NEVER independent biological replication of Role B
- 7 NEVER uniform benefit across the six conditions
- 8 NEVER a confirmed Role-A result
- 9 NEVER single-cell input equivalence, the model being trained on clone pseudobulk

The result is bounded to one cell line, six observed experimental conditions, and one observed detection proxy. It is not evidence about therapy, and the six conditions include non-clinical stress contexts that no one would administer to anything.

Generation 2

What would actually test this, none of which was a gate on the present result:

independent biological replication in a different lineage-traced system
transfer to conditions the model has never seen, which requires a condition
representation the present design does not have
a dataset-independent nuisance definition, since the current abundance block counts
cells in this experiment's three specific pretreatment libraries and therefore
cannot be computed anywhere else
calibration, frozen and tested separately
out-of-distribution behaviour

The first is the one that matters. Everything here rests on a single lineage-traced system, and one system is one system however carefully it is evaluated.

Conclusions

In the WM989 lineage-traced melanoma system, a clone's gene expression before treatment carries information about which of six observed experimental conditions it is still detected after: information specific to each condition, beyond condition identity and captured pretreatment clone abundance, under clone-held-out evaluation. Under a test preregistered in full, all of the gain in ordering a clone's six conditions came from an explicit state-by-condition interaction; an additive state term contributed nothing.

What this means is that pre-existing state in this system is not only a matter of how detectable a clone is overall. Part of it bears on which condition the clone is still detected after. A state contribution shared additively across all conditions cannot change their ordering within a clone, so an analysis that admits state only that way cannot see this part, as the additive model here could not; allowing state effects to vary by condition improved the ordering. That does not mean separate biology behind each condition: one programme whose effect differs between conditions would show the same pattern.

This opens directions others can take. The interaction terms of the frozen model are a starting point for finding the pretreatment programmes behind these condition-dependent effects, whether one or several. Any lineage-traced experiment that splits clones across several perturbations can ask the same question with the same design: ordering within a clone, abundance held fixed in every model, a null that refits the whole pipeline, and a protocol fixed before the statistic is computed. The most informative of these is a test in an independent lineage-traced system, which has not yet been done. If the result holds there, pretreatment profiling could help choose which perturbations to test on which subpopulations in experimental work. The outcome measured here is a detection proxy, not death or clinical response; that is where the result points, not what it shows.

Everything needed to reproduce the result from the public data, and to apply the design to new data, is released with this work.

List of abbreviations

- **AUROC** — area under the receiver operating characteristic curve
- **B** — the captured-abundance nuisance block
- **C1, C2** — the two endpoint families: post-treatment clone detection, and clone-balanced abundance
- **CI** — confidence interval
- **CoCl₂** — cobalt(II) chloride
- **CP10K** — counts per 10,000
- **delta_RANK** — W5 minus W4 in equal-clone-weighted within-clone AUROC
- **delta_TOP1** — W5 minus W4 in the lowest-predicted-score top-choice diagnostic
- **GEO** — Gene Expression Omnibus
- **PCA** — principal component analysis
- **U** — the condition-identity indicators
- **W1, W4, W5** — the three preregistered model specifications
- **X** — the clone expression profile, as 50 train-only principal components

Declarations

Ethics approval and consent to participate

Not applicable. This study is a computational reanalysis of published, publicly available data from an immortalised cell line. No human participants, human material or animals were involved.

Consent for publication

Not applicable.

Availability of data and materials

The dataset analysed here is publicly available from the Gene Expression Omnibus under accession GSE279162, generated and deposited by Schaff et al. [1].

Supporting Role-A evidence uses GSE227151. **No new data were generated for this study.**

All analysis code, frozen protocols, stage records, out-of-fold predictions, the serialized predictor and the verification tooling are archived at Zenodo, DOI [10.5281/zenodo.22769563](https://doi.org/10.5281/zenodo.22769563), and developed openly at <https://github.com/hagai-tty/Cellular-Outcomes-of-Perturbations>.

Verify before reading anything else

```
python experiments/run_gen1_evidence_lock.py --verify
python experiments/run_gen1_claim_lock.py --verify
python experiments/run_gen1_manuscript.py --verify
```

The first re-hashes every locked artifact and refuses if one has moved. The second does the same for the claim set. Both were shown to refuse a one-bit change before either was issued. The third does the same for this manuscript and the package around it. All three also refuse if the stage that produced the files did not pass.

```
evidence lock digest    60602531449079b8f86debea9ffd69753f8bfad033be4c476751d39da08c78aa
claim lock digest      fc6dc2f221acf1c7661e36071b9e377571c8f903d38398b55bedc164db7efa61
```

Licensing

```
software and frozen model  PolyForm Noncommercial License 1.0.0
                           SPDX: PolyForm-Noncommercial-1.0.0
manuscript text and figures CC BY 4.0
GSE279162, GSE227151      original depositors' terms; NOT relicensed here
```

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What is in the repository

Benchmark tables, frozen out-of-fold predictions, the serialized model metadata, the prediction API and CLI, the model card and schema, every stage protocol, every stage record, and every executor and contract file. Full inventory and per-file hashes: `results/evidence_lock/`.

What is not

stage24_w5_artifact.npz	44 MB, gitignored. A fresh clone does NOT contain it. Its hash is locked and it rebuilds in about half a minute: python experiments/run_stage24_gen1_tool.py --stage 24c The Zenodo archive does include it, so an unpacked archive verifies without a rebuild.
raw sequencing data	GSE279162, GSE227151. Not vendored. The size and SHA-256 of every input file are recorded, and the pipeline rebuilds from the public files, refusing on any byte that differs.

Neither is a gap any longer. The artifact is one command away, and a fresh clone rebuilt the clone pseudobulk and the model byte for byte from the public files and recovered the observed statistic exactly (REPRODUCIBILITY.md §2.1).

Full reproduction instructions, environment and runtimes:

results/manuscript/REPRODUCIBILITY.md.

Competing interests

The author holds the copyright in CellFate-Rx and offers commercial licences for it. The author has received no income from it to date.

Funding

This work received no external funding.

Authors' contributions

H.A. conceived the study, set its logic and direction, and directed the AI-assisted implementation and documentation described in Methods. H.A. reviewed the work and approved the final manuscript.

Acknowledgements

We thank Schaff et al. for generating and openly depositing GSE279162, without which this reanalysis would not be possible.

Authors' information

Not applicable.

References

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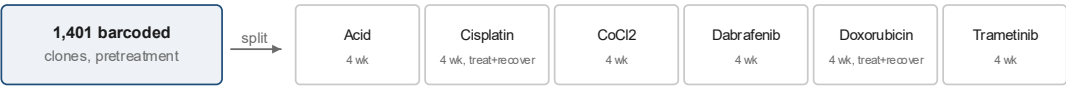
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Figure legends

Figures are generated from the locked result files by python
experiments/make_gen1_figures.py; no number in them is typed by hand.

Figure 1 Clone-level prospective design and evaluable population

One BRAF-V600E melanoma line (WM989). Data from Schaff et al. 2026; reanalysis only.

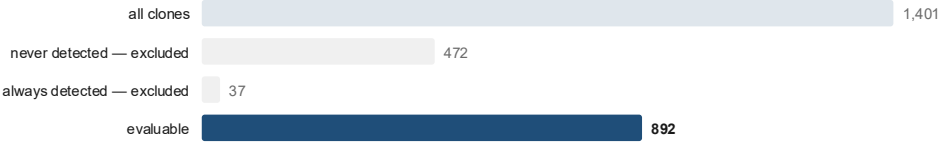


Outcome C1 post-treatment clone DETECTION — an observed zero means no assigned cell was seen.

Not death, not sensitivity, not clinical response.

Evaluable population

A within-clone AUROC is undefined without both a detected and an undetected condition.



Closed vocabulary

56 of 56 adversarial condition strings refused, including 16 oncology drugs (Vemurafenib, Carboplatin, ...).

Design width 309 columns = 50 PCs + 4 abundance + 5 condition + 250 interaction.

Figure 1. Clone-level prospective design and evaluable population. The design in WM989, and the funnel from all clones to the evaluable subset.

Figure 2 Preregistered clone-specific ranking result
Equal-clone-weighted within-clone AUROC. Every degree of freedom fixed before these numbers existed.

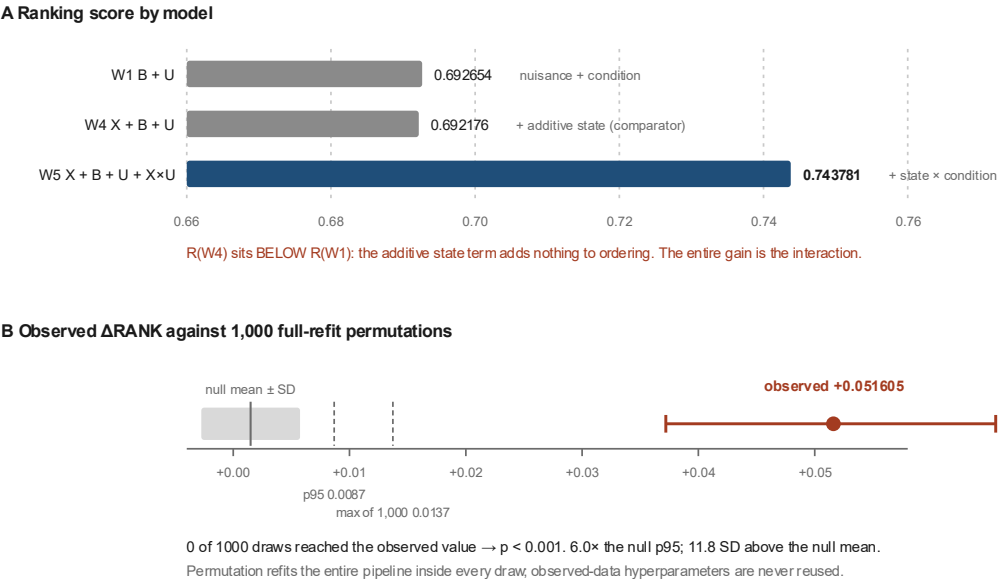


Figure 2. Preregistered clone-specific ranking result. (A) Ranking score by model: W1, W4 and W5. **(B)** The observed Δ RANK against 1,000 full-refit permutations.

Figure 3 Robustness across strata, and the top-choice diagnostic
Preregistered as descriptive: these could withhold support, never grant it.

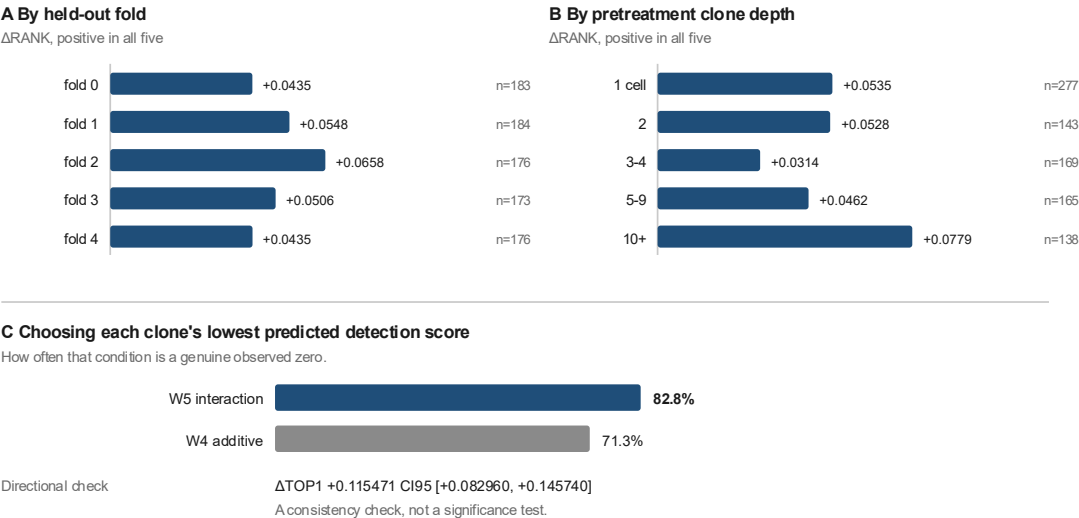


Figure 3. Robustness across strata, and the top-choice diagnostic. (A) Δ RANK by held-out fold. **(B)** Δ RANK by pretreatment clone depth. **(C)** Choosing each clone's lowest predicted detection score under W5 and W4 — a consistency check, not a significance test.